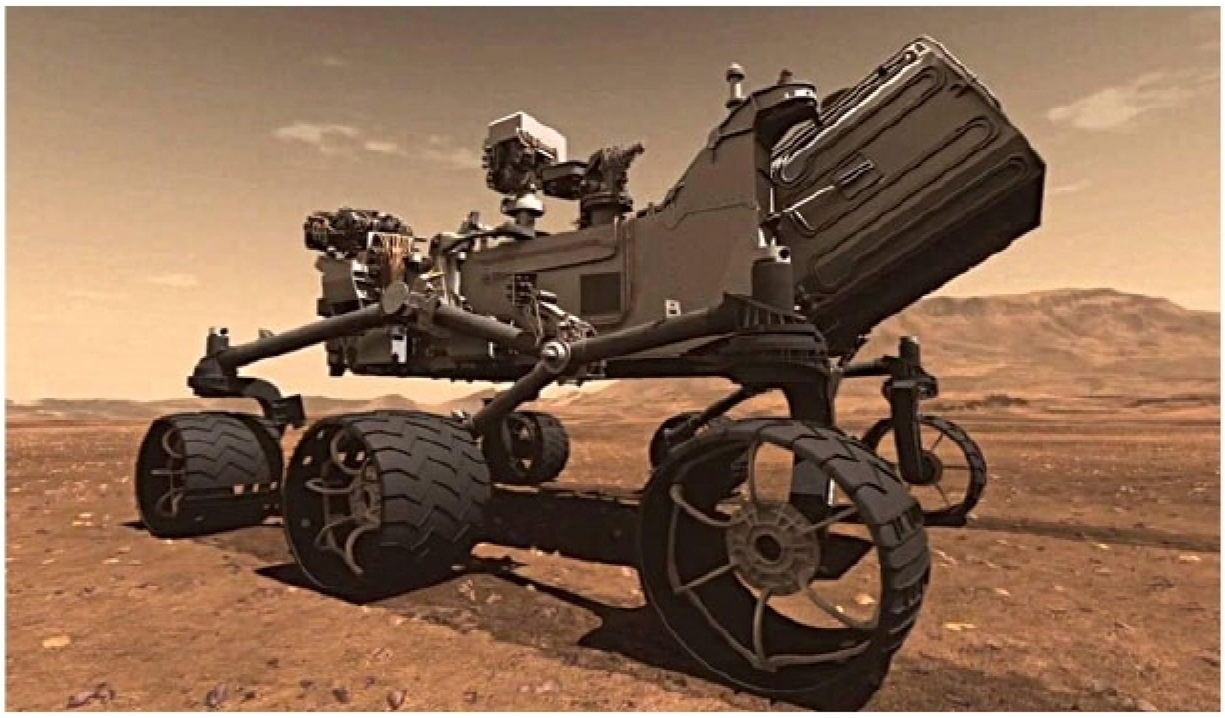
(7 feet). It weighs 900 kilograms. It has many cameras, rock vapouriser (laser-operated), rocker-bogie suspension and geology laboratory. The main mission of Curiosity was to find whether Mars can support small life forms called microbes. Curiosity was designed to travel up to 200 metres per day as it explored Mars.

Mars exploration rover (MER) charge-coupled device (CCD) sensors are mounted on it at two different types of cameras. These are hazard avoidance cameras (hazcams) and navigation cameras (navcams). Together these work to provide a complementary and comprehensive view of the terrain. Eight black-and-white hazcams are mounted on the lower portion of the front and back of the rover.

Other instruments that Curiosity carries are rock abrasion tool, alpha particle x-ray spectrometer, x-ray diffraction meter, neutron spectrometer used to measure hydrogen, ice or water, rover environmental monitoring system that consists of various sensors (including UV, air temperature, wind, ground temperature, humidity and pressure sensor), sample analysis at Mars (SAM) system that includes quadrupole mass spectrometer to detect gases, gas chromatograph used to separate individual gases from a complex mixture into molecular components and tunable laser spectrometer (TLS) that performs precision measurements of oxygen and carbon isotopes, ratios of

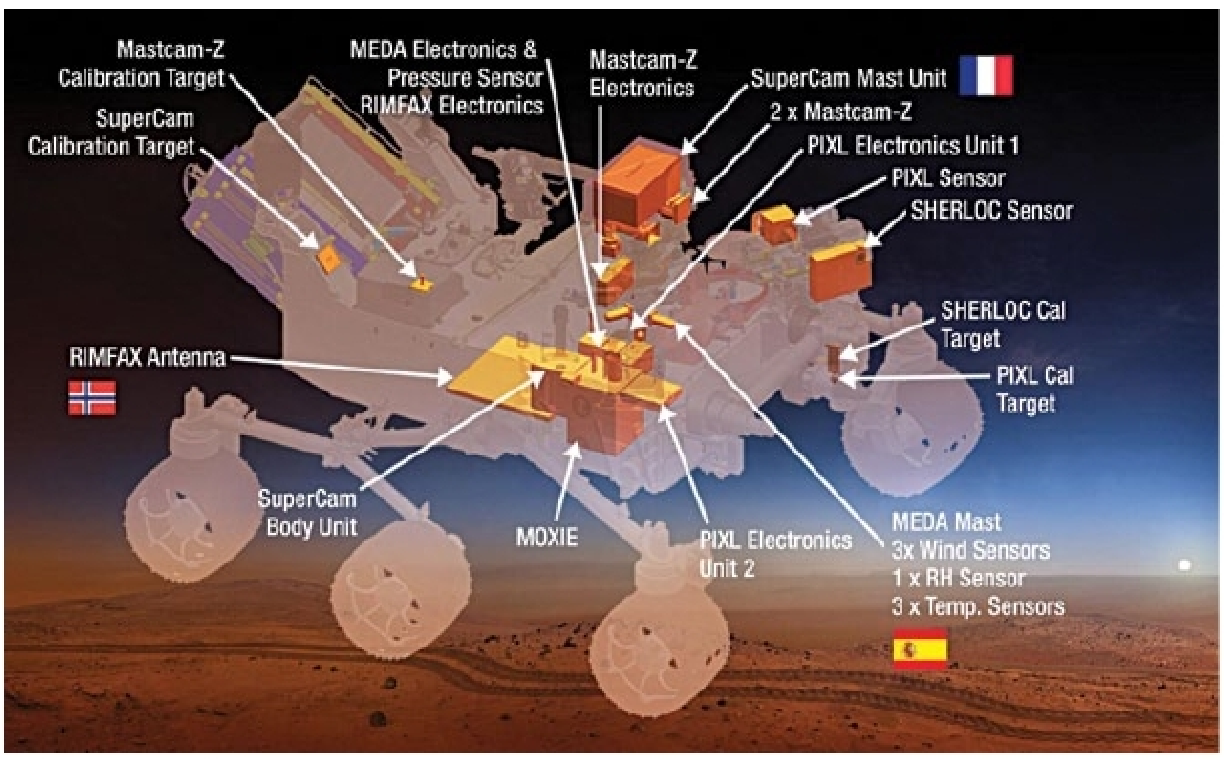


NASA's Curiosity rover on Mars

carbon-dioxide and methane in the atmosphere to distinguish between their geochemical or biological origin, and radiation assessment detector.

On March 2018, Curiosity celebrated 2000 days on Mars, making its travel from Gale Crater to Aeolis Mons. During the travel, it examined geological information embedded in mountains' layers, and found evidence of past water and geological change.

NASA's deep space network (DSN) is an international network of antennae that provides communication links between scientists and engineers and rovers on Mars. The DSN consists of three deep space communication facilities placed approximately 120-degrees apart around the world at Gold Stone in California's Mojave Desert, Real Madrid Spain and near Canberra, Australia.



NASA's proposed Mars 2020 rover, Maven

Curiosity has its own curiosities. One of them is its laser detection system that works with new artificial intelligence (AI)-based AEGIS algorithm, which took ten years to build and test.

In nutshell

Hardware-based features of rovers such as sensors, cameras, spectrometers and detectors should be strong and tamper-proof to face such hazardous elements as wind, storm, radiation, temperature, pressure, etc, so that uninterrupted data can be transmitted and collected by these devices and sent to Earth. Memories and processing capacities of brain (computer) of the rover should be strong, too.

Scientists and engineers are accepting rovers as the most essential part in planet exploration with more features and updated processes. More functionality can be added to rovers and correctly implemented. The present challenging task for researchers is to add nanotechnology, AI, Big Data, robotics, improved computer vision and other recent technologies to rovers to make them more sophisticated so that these gather better information required to make space life possible for humans.

Currently, landing of humans on Mars is the biggest challenge before scientists. NASA has been working on what it calls a Mars rover concept vehicle. It is a human-driven rover. NASA is hopeful of making astronauts land on Mars along with the rover by 2030. EFY